Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 7 NeuroFleetX – Urban Power Fleet Management System

Start date : 10-11-2025

Names: Hansika Gurjar,

Saisree,Prashant Kandnekar

Internship Duration: 8 Weeks

1. Project Title

NeuroFleetX – Urban Power Fleet Management System

2. Project Objective

NeuroFleetX is an AI-driven fleet management system designed to fully digitize and automate urban public transportation. By leveraging real-time data and smart automation, the project aims to replace rigid, manual transit systems with a dynamic digital ecosystem.

2.1 Centralized Fleet and Resource Management

- Unified Operational Control: To develop a centralized command center where administrators can monitor the real-time status (on-duty, maintenance, or idle) of every bus in the fleet.
- Driver & Route Optimization: To empower admins to dynamically assign drivers to specific routes based on real-time requirements, ensuring optimal utilization of human resources and preventing service gaps.
- Inventory Digitization: To eliminate manual paper-based logs by migrating all bus details, driver profiles, and maintenance records to a secure cloud-based database.

2.2 Real-Time Passenger Empowerment

- Eliminating Commuter Uncertainty: To provide passengers with live GPS-based tracking, significantly reducing the waiting time at bus stops.
- Accurate ETA Prediction: To calculate and display precise Estimated Time of Arrival (ETA) based on current traffic patterns and distance, allowing commuters to plan their journeys efficiently.

- Universal Digital Accessibility: To ensure that critical transit information—including bus timings, fare structures, and route maps—is easily accessible via any smartphone.

2.3 Implementation of AI and Smart Assistance

- 24/7 Virtual Support (NeuroBot): To deploy an AI-powered Chatbot capable of providing instant, automated responses to passenger queries regarding schedules and fares, eliminating the need for manual helpdesks.
- Data-Driven Decision Making: To analyze system data to identify high-traffic routes and peak hours, enabling authorities to allocate additional buses where they are needed most.

2.4 Seamless Ticketing and Revenue Management

- Paperless Transactions: To replace traditional cash-based ticketing with a digital solution, enhancing transparency and solving the common issue of managing small currency changes.
- Automated Revenue Tracking: To provide administrators with accurate, daily reports on ticket sales and revenue generation through an automated financial dashboard.

2.5 Scalability and Sustainability for Smart Cities

- Future-Ready Architecture: To design a modular system that is scalable from a small town operation to a full-scale metropolitan transit network.
- Environmental Sustainability: To optimize bus routes to reduce unnecessary mileage and fuel consumption, promoting eco-friendly public transportation as a viable alternative to private vehicles.

3. Project description in detail

NeuroFleetX is a sophisticated, enterprise-grade **Urban Power Fleet Management System** engineered to revolutionize the infrastructure of public transportation. This project is built on a high-performance **Full-Stack Architecture**, combining the dynamic frontend capabilities of **React.js** with the robust, secure, and scalable backend processing power of **Java Spring Boot**.

The system is designed to transition traditional transit operations from manual, fragmented workflows into an intelligent, automated, and data-driven digital ecosystem.

3.1 Technical Architecture

NeuroFleetX utilizes a modern, service-oriented architecture to ensure high availability and performance:

- **Frontend (The Interface Layer):** Developed using **React.js** and **Tailwind CSS**, the frontend provides a highly responsive and intuitive dashboard. It features dedicated portals for Admins, Drivers, and Passengers, ensuring that real-time data is presented clearly across all devices.

- **Backend (The Logic Layer):** Powered by **Java Spring Boot**, the backend serves as the core engine. It manages complex business logic, handles heavy concurrent requests, and provides enterprise-level security through **Spring Security** and **JWT (JSON Web Tokens)** for role-based access control.

- **Database (The Data Layer):** Utilizing **PostgreSQL/MySQL**, the system maintains data integrity for fleet records, driver shifts, and passenger transaction history.

3.2 Core Functional Modules

- **Real-Time Fleet & Route Intelligence:** Leveraging the multi-threading capabilities of Spring Boot, the system tracks the live location of every vehicle in the fleet. The Admin Dashboard provides a comprehensive visualization of the city's transit health, showing active routes, vehicle status, and operational bottlenecks.

- **AI-Powered Passenger Support:** The system features an integrated **AI Chatbot (NeuroBot)**. By connecting Spring Boot APIs with advanced Generative AI models, the chatbot provides instantaneous support to commuters regarding schedules, fares, and route changes, simulating a human-like assistance experience.

- **Automated Scheduling & Resource Allocation:** The backend logic optimizes the assignment of drivers and buses to specific routes. This ensures that transport resources are deployed efficiently during peak hours, reducing commuter wait times and operational costs.

- **Secure Digital Ticketing & Analytics:** NeuroFleetX includes a seamless ticketing module where passengers can book digital passes. Spring Boot's robust transaction management ensures that all financial data is processed securely, while the integrated analytics engine generates visual reports on revenue and passenger flow.

3.3 Enterprise Advantages of Spring Boot

- **Scalability:** The choice of Spring Boot allows NeuroFleetX to scale effortlessly, supporting thousands of concurrent users and vehicles without compromising performance.

- **Robust Security:** Implementation of Spring Security ensures that sensitive operational data and user information are protected against unauthorized access.

- **Reliability:** With Spring Data JPA and automated transaction management, the system ensures 100% data consistency, which is critical for fleet logistics and financial tracking.

3. Timeline Overview

The development of NeuroFleetX was executed over an 8-week intensive cycle, following an Agile methodology to ensure seamless integration between the React.js frontend and the Spring Boot backend.

Week	Activities Planned	Activities Completed
Week 1	Requirement gathering, system study, and defining	Project scope finalized; identified key stakeholders

Week	Activities Planned	Activities Completed
	project scope for urban transit.	(Admin, Driver, Passenger).
Week 2	Tech stack finalization and environment setup for React and Spring Boot.	Development environment configured; **Spring Tool Suite (STS)** and **VS Code** set up.
Week 3	Database schema design and Backend initialization.	Designed ER diagrams; set up **PostgreSQL** database; initialized Spring Boot project.
Week 4	Development of Core Backend APIs and Security implementation.	Developed REST APIs for Bus and Route management; implemented **Spring Security & JWT**.
Week 5	Frontend UI Development and Dashboard wireframing.	Built React components for Admin and Passenger portals; integrated **Tailwind CSS**.
Week 6	Integration of AI Chatbot and Real-time data logic.	Integrated GenAI API with Spring Boot; developed logic for Live Tracking and ETA simulation.
Week 7	System Integration and End-to-End API Testing.	Linked React frontend with Spring Boot backend; performed API testing using **Postman**.
Week 8	Bug fixing, Documentation, and Final Deployment.	Conducted UAT (User Acceptance Testing); completed project report and deployed the application.

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Defined project objectives, finalized the use of **Spring Boot** and **React**, and assigned team roles.	26– Nov-2025
Prototype/First Draft	Developed the initial prototype featuring the core **Spring Boot API** structure and basic frontend layout.	09 - Dec-2025
Mid-Term Review	Completed the mid-term review to evaluate **JWT Security** implementation and Backend-Frontend integration.	22 - Dec - 2025
Final Submission	Submitted the completed **NeuroFleetX** project with a fully functional Monorepo and detailed documentation.	2 - Jan - 2025
Presentation	Delivered a comprehensive project presentation showcasing real-time tracking, AI chatbot, and Admin features.	6- Jan - 2025

5b. Project execution details

The execution of **NeuroFleetX** followed a structured Software Development Life Cycle (SDLC) to ensure the robustness of the enterprise backend and the responsiveness of the frontend.

- **Requirement Analysis:** Conducted a comprehensive study of urban transit challenges; defined the three-actor scope (Admin, Driver, Passenger) and finalized the automation objectives.
- **Technology Selection:** Opted for **Java Spring Boot** for a scalable backend, **React.js** for a dynamic UI, **PostgreSQL** for relationnal data integrity, and **Spring Security** for enterprise-grade protection.

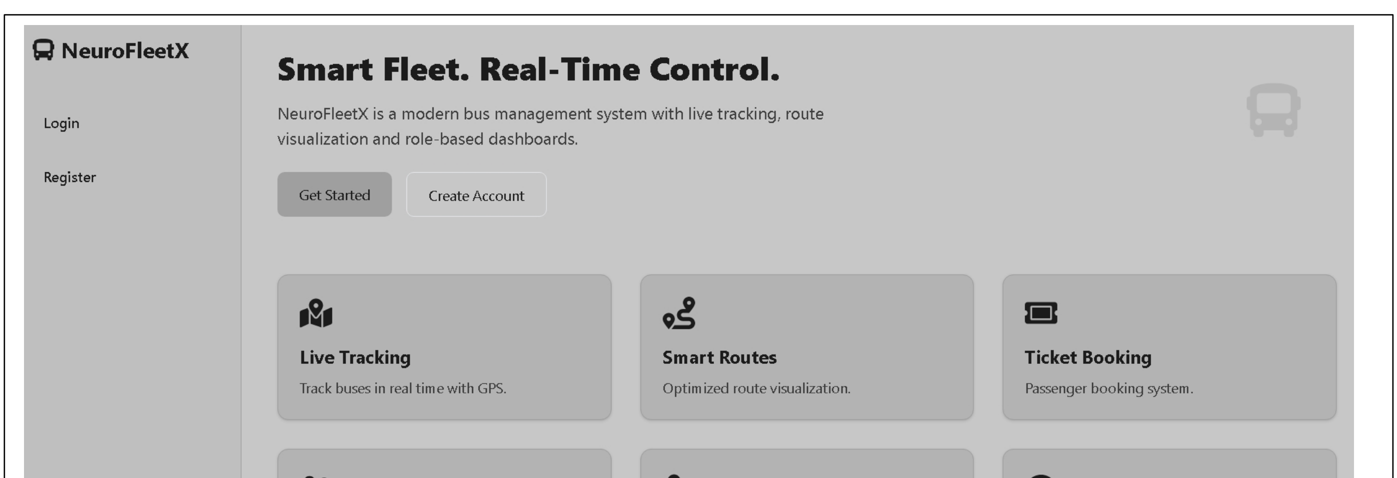
- **System Architecture & Design:** Designed the **Monorepo** structure, finalized the RESTful API endpoints, created ER diagrams for database relations, and designed UI wireframes for the Admin Dashboard.
- **Database Implementation:** Developed the schema in the database and utilized **Spring Data JPA** for Object-Relational Mapping (ORM) to handle complex bus and route entities.
- **Backend Development:** Engineered the core logic using Spring Boot; implemented **JWT-based authentication** and developed services for real-time fleet status updates and ticketing logic.
- **Frontend Development:** Created modular React components using **Tailwind CSS**; implemented interactive maps for bus tracking and a sleek interface for the AI Chatbot.
- **AI Integration:** Developed an integration layer between the Spring Boot backend and Generative AI APIs to power the **NeuroBot** virtual assistant for automated passenger support.
- **System Integration:** Connected the React frontend with Spring Boot REST APIs using Axios; ensured real-time data flow between the driver's status updates and the passenger's live view.
- **Testing & Quality Assurance:** Conducted extensive **API testing using Postman**, performed Unit Testing for backend services, and executed User Acceptance Testing (UAT) to ensure a bug-free experience.
- **Deployment & Documentation:** Finalized the project documentation, organized the GitHub repository, prepared the system installation guide, and successfully delivered the final project demonstration.

6. Snapshots / Screenshots

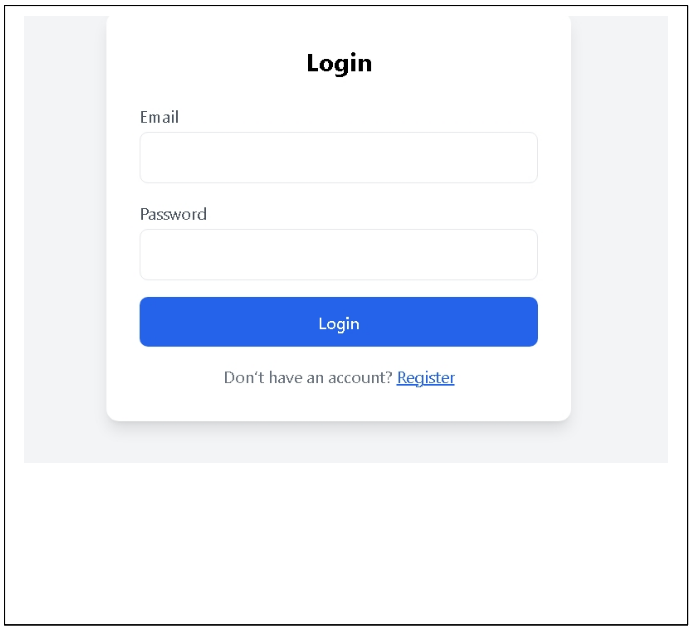
Step 1 : Go to Github and clone the Repository in your system then open in VS Code to do the execution , here is the github repo link :

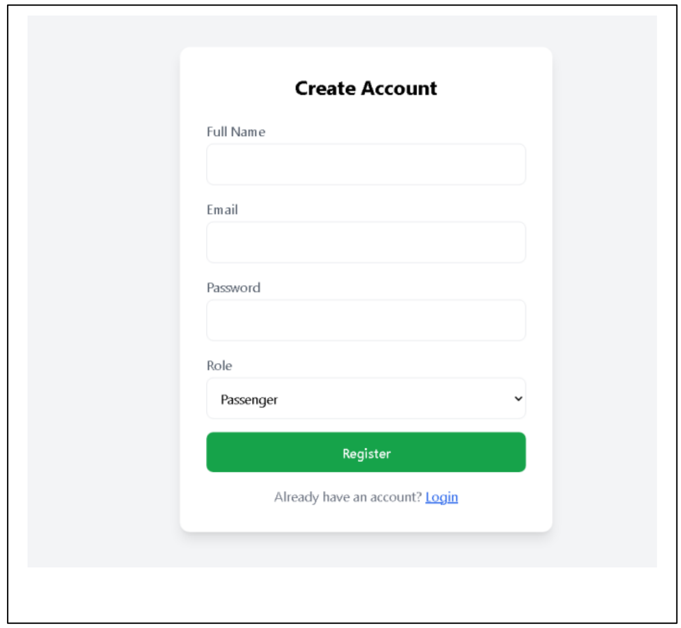
https://github.com/hansika-25/NeuroFleetX

Here is the Landing page with basic navigation

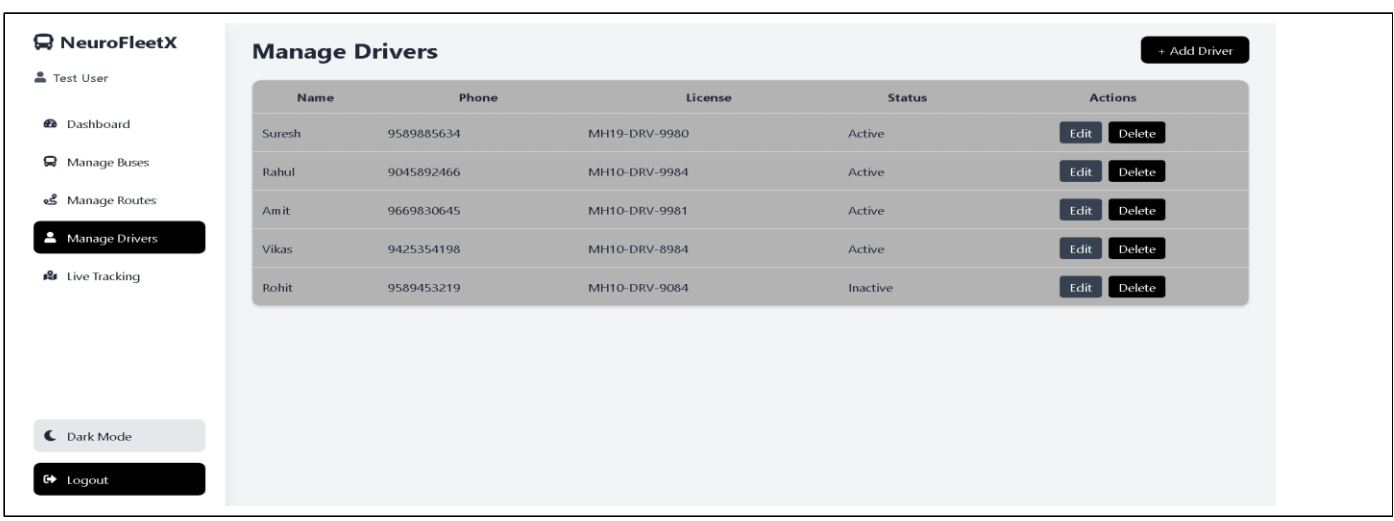


After clicking on Login here user need to register and log in for Authentication based on the role and also can create an account by clicking on the register.

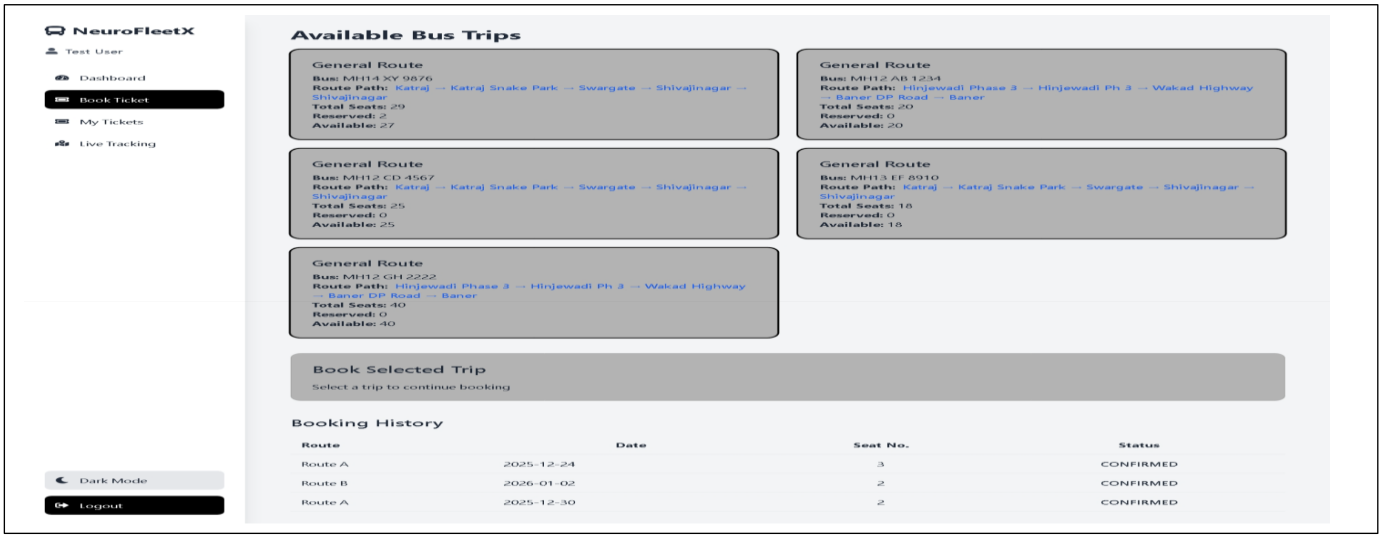




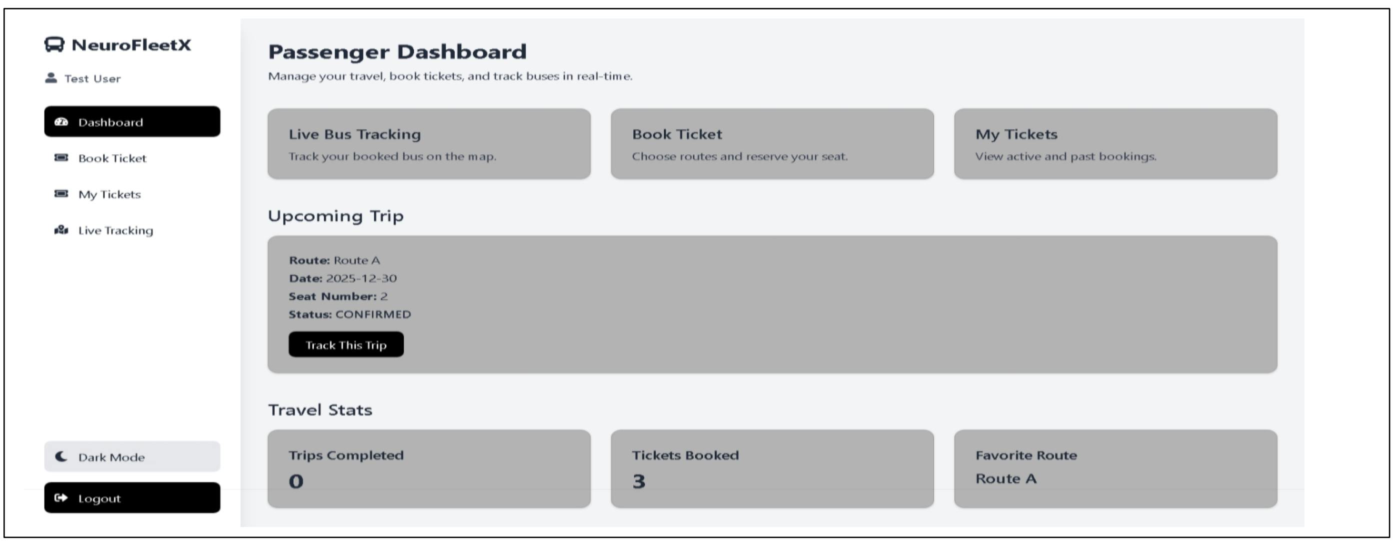
On the admin page the Admin can manage the drivers,routes and buses by adding, deleting and editing them



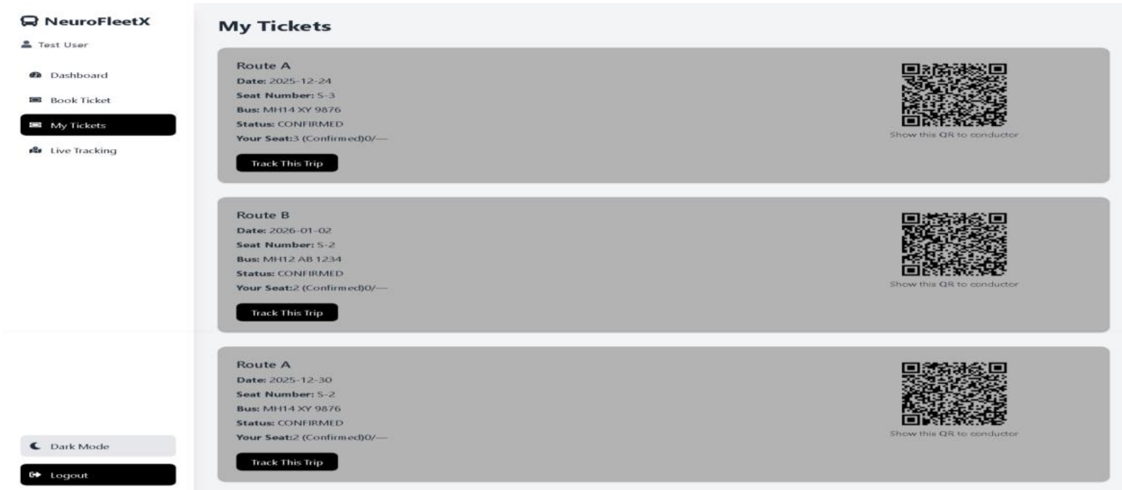
: On the passenger Dashboard the user can book the tickets.



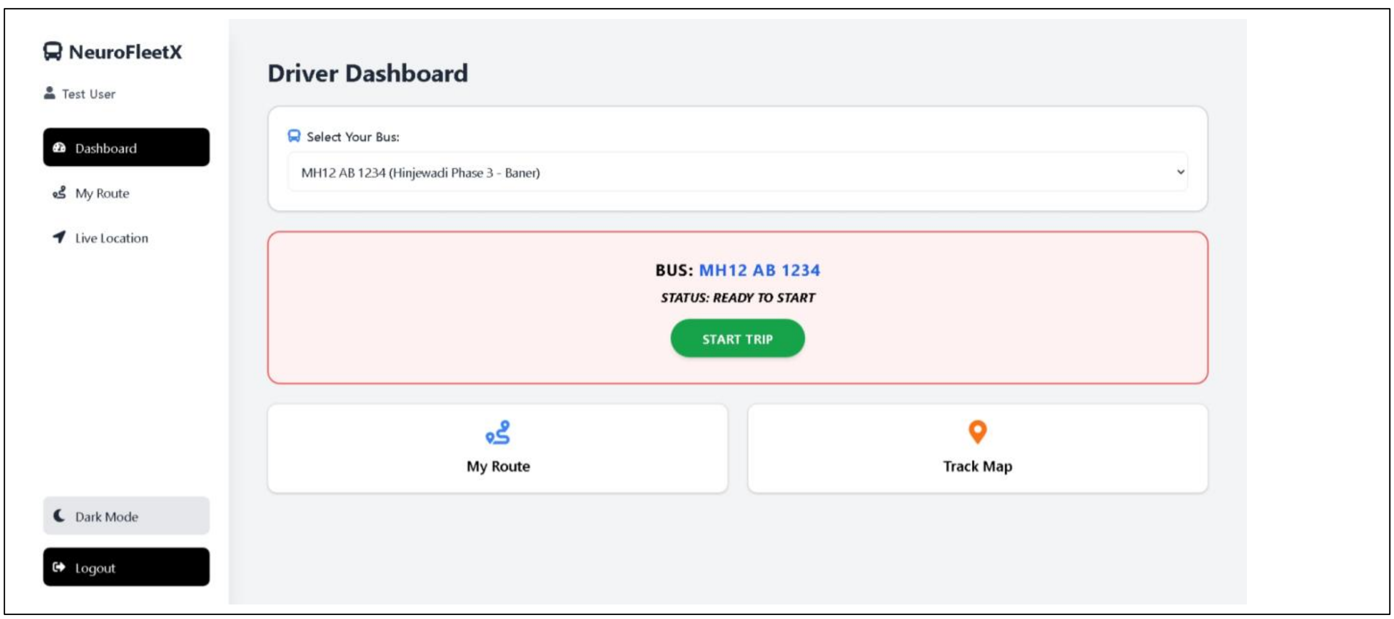
The user can see the upcoming trip on the passenger Dashboard



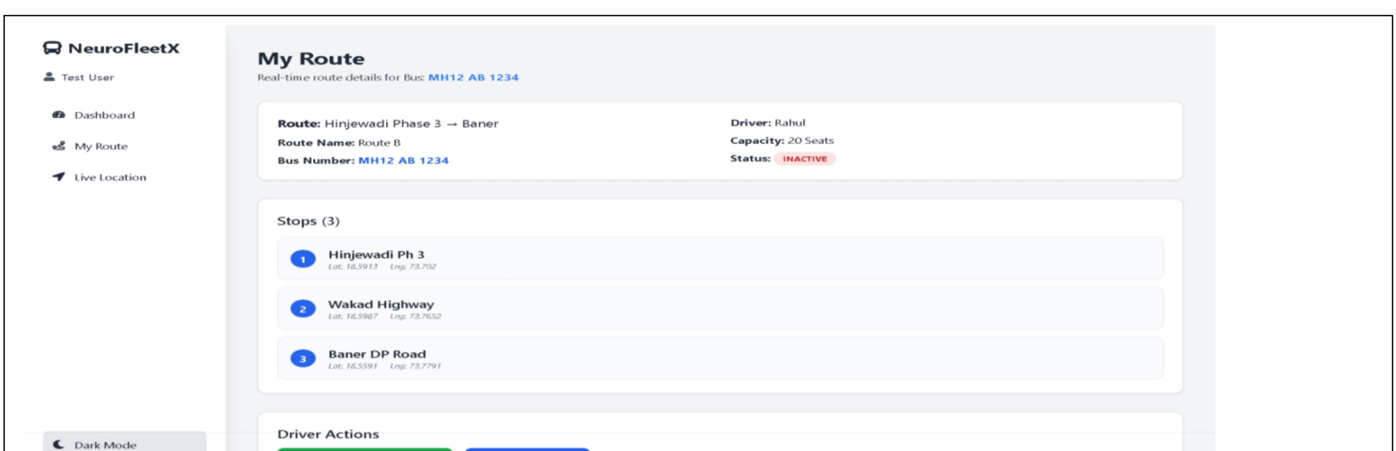
Here the user can see the tickets that he booked



this is the driver dashboard where the driver can select the Bus he wants to continue with and can start the trip



After that the user can also click on my route to examine their current route



7. Challenges Faced

- **System Integration Complexity:** Linking the **Spring Boot** REST APIs with the **React** frontend while managing a **Monorepo** structure caused initial synchronization and routing issues.

- **Real-Time Data Synchronization:** Ensuring that the bus location and status updated by the **Driver** were instantly visible to the **Passenger** and **Admin** without performance lag required optimizing API polling and state management.

- **Security & Role-Based Access:** Implementing robust security using **JWT (JSON Web Tokens)** to ensure that sensitive Admin data (revenue/driver info) remained inaccessible to unauthorized users was a complex task.

- **AI Chatbot Accuracy:** Fine-tuning the **NeuroBot** to provide precise transit information (like route timings and fares) required careful prompt engineering and backend integration with the GenAI API.

- **Handling Concurrent Requests:** Ensuring the **Spring Boot** backend could handle multiple requests from different stakeholders (Admin, Driver, and Passenger) simultaneously without database deadlocks.

- **Frontend Data Rendering:** Mapping complex JSON responses from the backend to the React UI, especially for dynamic lists like "Live Routes" and "Active Buses," required safe rendering and conditional mapping.

- **Database Schema Evolution:** Designing a relational database in **PostgreSQL** that could efficiently link Buses, Drivers, and Routes while maintaining data integrity was a significant design challenge.

- **Version Control & Branch Management:** Resolving merge conflicts in the Monorepo, especially when frontend and backend changes were pushed simultaneously to the same repository.

- **User Interface Design:** Creating a professional, mobile-responsive dashboard using **Tailwind CSS** that remained intuitive for non-technical users (Drivers and Passengers) required multiple iterations.

- **Time Management:** Balancing the development of an enterprise-level Java backend and a modern React frontend within the strict **8-week internship timeline** was demanding.

8. Learnings & Skills Acquired

- **Enterprise Backend Development:** Mastered the use of **Java Spring Boot** to build scalable, high-performance REST APIs and managed complex business logic for fleet operations.

- **Modern Frontend Engineering:** Enhanced expertise in **React.js** and **Tailwind CSS**, focusing on building dynamic, responsive dashboards that handle real-time data updates.

- **Security & Authentication:** Gained hands-on experience implementing **Spring Security** and **JWT (JSON Web Tokens)** to manage role-based access control (RBAC) for Admins, Drivers, and Passengers.

- **Database Management & ORM:** Improved skills in relational database design using **PostgreSQL/MySQL** and utilized **Spring Data JPA** for efficient object-relational mapping and data persistence.

- **AI Integration & Prompt Engineering:** Learned to integrate Generative AI capabilities into a full-stack application, enabling the **NeuroBot** chatbot to provide intelligent, automated passenger support.

- **Real-Time Data Synchronization:** Developed techniques for synchronizing data between the backend and frontend, ensuring that bus locations and ETAs remain accurate across all user interfaces.

- **Full-Stack Architecture & Monorepos:** Understood the complexities of managing a **Monorepo** structure, ensuring seamless coordination between frontend and backend codebases.

- **System Testing & Quality Assurance:** Strengthened debugging skills by performing extensive **API testing with Postman** and conducting end-to-end integration testing to ensure system reliability.

- **Agile Methodology & Project Management:** Adopted Agile practices, including sprint planning and iterative development, to complete the project within the 8-week internship timeline.

- **Domain Knowledge in Transit Logistics:** Developed a deep understanding of urban transportation challenges, including route optimization, fleet scheduling, and digital ticketing systems.

9. Testimonials from team

Team Member 1: "Developing NeuroFleetX significantly deepened my understanding of building enterprise-grade backends using **Java Spring Boot**. I gained hands-on experience in implementing **JWT-based security** and managing complex relational data with **Spring Data JPA**, which are essential skills for developing robust, real-world applications."

Team Member 2: "Working on this project helped me master **React.js** for creating dynamic and responsive user interfaces. I learned how to handle real-time data synchronization between the frontend and the backend, and how to use **Tailwind CSS** to build a professional-grade dashboard for fleet and route management."

Team Member 3: "NeuroFleetX was a great learning curve in **AI integration and Full-Stack coordination**. I learned how to bridge the gap between Generative AI APIs and a Spring Boot backend to create the **NeuroBot** assistant. Additionally, managing a **Monorepo** structure taught me the importance of version control and collaborative project management."

10. Conclusion

The development of **NeuroFleetX** has been a comprehensive journey through the modern software development lifecycle. By successfully integrating a high-performance **Spring Boot** backend with a dynamic **React** frontend, the project addresses critical gaps in urban transit management.

The implementation of real-time tracking, secure role-based access, and AI-driven passenger support demonstrates the potential of full-stack technologies in building "Smart City" solutions. This project not only delivered a functional tool for fleet management but also equipped the team with industry-standard technical skills and problem-solving abilities.

11. Acknowledgements

I would like to express my sincere gratitude to Infosys for providing me with the opportunity to work on the NeuroFleetX: Urban Power Fleet Management System project and gain hands-on experience in full-stack development and AI integration.

I am especially thankful to my mentor, Sangeetha, for her constant guidance, support, and valuable insights throughout the internship. Her mentorship was instrumental in helping me master the complexities of the Spring Boot framework, implement secure JWT authentication, and overcome technical challenges during the development process. Her feedback helped me grow both technically and professionally.

I would also like to thank my team members for their constant collaboration, encouragement, and dedication toward successfully executing this project. Our collective efforts in integrating the React.js frontend with the backend and refining the NeuroBot AI made this internship journey smooth, productive, and enriching.

Finally, I appreciate everyone who contributed directly or indirectly to my learning and the successful completion of this project.